Lecture 6 (*March 1*)

Reading: Section 3.1
Viewing: Lectures 5-6

A key idea of linear algebra is to think of a line/plane/space as a **vector space**. Specifically, this means that points of n–dimensional space:

$$\mathbb{R}^n = \text{set of all } n\text{–tuples of real numbers} = \Big\{(x_1, \ldots, x_n),\ x_1, \ldots, x_n \text{ are real numbers}\Big\} \tag{48}$$

can be thought of as column vectors, according to the correspondence:

$$(x_1, \ldots, x_n) \quad \leftrightarrow \quad \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

For example, $\mathbb{R}$ is the line, $\mathbb{R}^2$ is the plane, $\mathbb{R}^3$ is usual space. But the key thing about thinking of these as sets of vectors is that vectors can be added, and vectors can be multiplied with scalars:

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix} \qquad \text{and} \qquad \alpha \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \vdots \\ \alpha x_n \end{bmatrix} \tag{49}$$

Definition 3. *A **vector space** is a set V, together with a notion of how to add two elements in V, and a notion of how to multiply an element of V by a scalar. In other words:*

$$\textit{for any } \boldsymbol{v}, \boldsymbol{w} \in V, \textit{ there must be a rule for defining } \boldsymbol{v} + \boldsymbol{w} \in V \tag{50}$$

$$\textit{for any } \boldsymbol{v} \in V, \alpha \in \mathbb{R}, \textit{there must be a rule for defining } \alpha\boldsymbol{v} \in V \tag{51}$$

This may seem abstract, but I will now spell out the main examples of vector spaces we will encounter in this course. First of all, $\mathbb{R}^n$ is a vector space, because of (49): the sum of vectors is a well-defined vector, and a vector times a scalar is a well-defined vector. More generally, we have:

Definition 4. *If V is a vector space, then a subset $S \subset V$ is called a **subspace** if:*

$$\textit{for any } \boldsymbol{v}, \boldsymbol{w} \in S, \textit{ then } \boldsymbol{v} + \boldsymbol{w} \in S \tag{52}$$

$$\textit{for any } \boldsymbol{v} \in S, \alpha \in \mathbb{R}, \textit{then } \alpha\boldsymbol{v} \in S \tag{53}$$

where the notions "$\boldsymbol{v} + \boldsymbol{w}$" and "$\alpha\boldsymbol{v}$" are well-defined, because S is a subset of a vector space V.

By combining (52) and (53), we have the more general fact for any subspace S of a vector space:

$$\text{for all } \boldsymbol{v}_1, \ldots, \boldsymbol{v}_k \in S \text{ and } \alpha_1, \ldots, \alpha_k \in \mathbb{R}$$

then:

$$\boxed{\alpha_1\boldsymbol{v}_1 + \cdots + \alpha_k\boldsymbol{v}_k \quad \text{lies in} \quad S} \tag{54}$$

For any choice of scalars $\alpha_1, \ldots, \alpha_k$, the vector in (54) will be called a **linear combination** of the vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$.

In this course, we will be interested in subspaces $S \subset \mathbb{R}^n$. There is a mind-boggling infinity of <u>subsets</u> of $\mathbb{R}^n$, but only a select few of them will turn out to be subspaces. In the case $n = 1$, there are only two subsets $S \subset \mathbb{R}$ which are subspaces: $S = \{0\}$ (the zero subspace) and $S = \mathbb{R}$ itself (the whole line). In the case $n = 2$, the subsets $S \subset \mathbb{R}^2$ which are subspaces are of three types:

- $S = \{0\}$ (the zero subspace)
- $S =$ any line passing through the origin
- $S = \mathbb{R}^2$ (the whole plane)

Indeed, if a subspace $S \subset \mathbb{R}^2$ contains any non-zero vector, then it contains the whole line passing though the origin and that vector, due to (53). And if S contains two non-zero vectors not on the same line, then it contains the entire plane, due to (54). Based on this pattern, you can probably guess that the subsets $S \subset \mathbb{R}^3$ which are subspaces are: the zero subspace, any line passing through the origin, any plane passing through the origin, and the whole space.

Now comes the point where we explain what all of this has to do with linear algebra, specifically with solving systems of equations like $A\boldsymbol{v} = \boldsymbol{b}$. We start from the following definition.

Definition 5. *The **column space** of an $m \times n$ matrix A is the subspace:*

$$C(A) \subset \mathbb{R}^m \tag{55}$$

spanned by the columns of A (the matrix A has n columns).

*Here and throughout our course, the subspace **spanned** by any vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ is defined to be the set of all linear combinations* (54) *of those vectors.*

Exercise: prove that the subspace spanned by any number of vectors in $\mathbb{R}^m$ is a subspace, in the sense of Definition 4. While you ponder this question, here is an explicit example of column space:

$$\text{if } A = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \end{bmatrix} \text{ then } C(A) = \left\{ \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \text{ for all scalars } \alpha \text{ and } \beta \right\}$$

So in other words, the column space in the example above is the plane traced out by the vectors $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$ in space. However:

$$\text{if } B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix} \text{ then } C(B) = \left\{ \alpha \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \beta \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \text{ for all scalars } \alpha \text{ and } \beta \right\} =$$

$$= \left\{ \gamma \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \text{ for all scalars } \gamma \right\} \tag{56}$$

so the column space of B is the line determined by the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. This is because the second column $\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$ is already on the line spanned by the first column. Taking linear combinations of vectors that lie on the same line passing through the origin will still remain on the given line:

$$\alpha \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \beta \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 2\beta \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = (\alpha + 2\beta) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

for any scalars α and β.

Fact 5. *A system of equations $A\boldsymbol{v} = \boldsymbol{b}$ has solutions if and only if $\boldsymbol{b} \in C(A)$.*

The fact above is simply a restatement of Fact 2. It is simply saying that if there is a solution $\boldsymbol{v}$ to the system $A\boldsymbol{v} = \boldsymbol{b}$, then (5) implies that $\boldsymbol{b}$ is a linear combination of the columns of A. So the column space of a matrix precisely consists of those vectors $\boldsymbol{b}$ which can appear as the right-hand sides of systems of linear equations $A\boldsymbol{v} = \boldsymbol{b}$.

For example, does the system:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{57}$$

have any solutions? The answer is no, because we have already seen in (56) that the column space of the matrix in question is the line spanned by the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Since the vector $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is not on this line (as it is not a multiple of $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$), then Fact 5 implies that the system (57) has no solutions.